

ADG classes: Minimizers

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Today we will use data from metagenomic samples collected during the scientific expedition Tara Oceans (2009-2012) from the South Atlantic Ocean. Seawater from different depths was filtered to retain small particle sizes (virus particles), small cell sizes (bacteria), or large cell sizes (protists). The DNA was extracted and submitted to high throughput sequencing. The files provided on Moodle contain 100 k filtered reads each; all reads are 100 nucleotides long, and contain no missing (N) nucleotides.

For hashing, we will use MurmurHash3 (`mmh3` Python package), you can install it in a new virtualenv as follows:

```
python3 -m venv .venv
source .venv/bin/activate
pip3 install mmh3
```

Hint: you may wish to work on a smaller subset of reads for most of the exercises below.

1 Unique k -mers.

1. Fix a value of k . How many unique k -mers does each of the samples contain? Compare the results to the number of unique k -mers in a set of random 100-mers of the same size as the samples.
2. What values of k would be reasonable to use?

Hint: nowadays most computers deal with 64-bit and 32-bit words, and a DNA nucleotide can be stored in 2 bits.

Another hint: many tools consider *canonical sequence* of a k -mer pair, i.e. lexicographically smaller of the two reverse complementary sequences. (Today, we can ignore the reverse complementation.)

2 Minimizers.

1. Inspect the naïve code for minimizer extraction.
2. Consider the 31,15-minimizer scheme for a selected sample. How many minimizers does an average read have?
3. Take all minimal 15-mers of the 31-mers found in the reads in the sample. Map these minimizers to integers using `lmer_to_int()`; these integers will sort in lexicographic order of k -mers. What is the distribution of these integer values?
4. What can be done to make the minimizer partitionings more uniform?
One idea: XOR the integer with some random value. Or with a carefully designed one, say 0b0011 repeated, i.e. flip every second nucleotide. Another idea: ban a subset of k -mers from being minimizers.